

Sagar Singh

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science, CGPA: 8.3/10

Vellore, India

2021 - 2025

EXPERIENCE

TranquilAI - Backend Developer

May 2025 - July 2026

Tech Stack: Java, Spring Boot, Spring Security, Spring Data JPA, Hibernate, PostgreSQL, JWT, OAuth 2.0, Kafka, Redis, Docker, Kubernetes, GitHub Actions, Maven, Git

- Developed 15+ RESTful APIs enabling secure authentication, AI conversations, journaling, and mood tracking for 500+ users.
- Improved security with JWT and OAuth 2.0, used multithreading for concurrent request handling, and achieved 80%+ test coverage with JUnit 5 and Mockito.
- Built Controller-Service-Repository architecture following SOLID principles, containerized services with Docker, orchestrated with Kubernetes, and reduced release time by about 40% using GitHub Actions CI/CD.
- Implemented Kafka-based async communication and Redis session caching/rate limiting, improving throughput by about 35%.

PROJECTS

Atlas - Microservices Backend Platform

Java 21, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, PostgreSQL, Kafka, Redis, Docker, Kubernetes, Maven, JWT

- Architecting a production-grade microservices platform with Auth, Workspace, Diagram, Deployment, Simulation, and AI Review services behind a central API Gateway.
- Implemented Auth Service with JWT authentication, BCrypt password hashing, DTO API boundaries, Bean Validation, JPA/Hibernate, and PostgreSQL constraints.
- Applied SOLID principles and domain-driven design so each service owns its data and business logic while shared modules remain scoped to contracts and utilities.
- Designed synchronous HTTP flows for fast operations and Kafka workflows for long-running deployment pipelines, with Docker Compose for local orchestration.

Student Management API

Java, Spring Boot, Spring Data JPA, Hibernate, PostgreSQL, Docker, JUnit 5, Swagger

- Developed a RESTful API with layered Controller-Service-Repository architecture, CRUD operations, pagination, sorting, and search.
- Enforced data integrity with Bean Validation and centralized exception handling, validated business logic with 10+ tests, and documented 20+ endpoints using Swagger/OpenAPI.

TECHNICAL SKILLS

Languages: Java, C/C++, JavaScript, SQL

Libraries/Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Spring Cloud, Spring Security, REST APIs, JUnit 5, Mockito

Tools / Platforms: Git, GitHub, Maven, Docker, Kubernetes, Postman, Swagger/OpenAPI, VS Code

Databases: PostgreSQL, MySQL, Redis

Concepts: SOLID Principles, Design Patterns, OOP, Microservices, Unit Testing, Integration Testing, Agile/Scrum, Domain-Driven Design

CERTIFICATIONS

AWS Certified Cloud Practitioner

Software Engineering Job Simulation: JPMorgan Chase & Co.